Waseem Raza, PhD

Faculty Candidate | Optimization for Machine Learning & AI-Enabled Networked Systems

Robust Learning | Reinforcement Learning | Network Optimization | Scientific / Domain-Informed ML

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LinkedIn | Google Scholar | ORCID | GitHub | Portfolio

Academic Profile

AI/ML researcher and engineering educator with a PhD in Electrical and Computer Engineering and research spanning optimization, reinforcement learning, robust machine learning, networked intelligent systems, 5G/6G, digital twins, and wearable intelligence. My work develops and rigorously evaluates data-driven methods for decision-making under limited data, distribution shift, heterogeneous environments, and operational uncertainty. Research experience combines analytical modeling, Python-based simulation, experimental testbeds, transfer learning, domain adaptation, generative modeling, statistical validation, and industry R&D. Published work includes multi-parameter optimization and proactive reliability, robust propagation models for digital twins, intelligent outage management, user-centric RAN optimization, and AI-enabled sensing.

Research Interests & Methodological Fit

Optimization for ML & AI-Enabled Decision Making: multi-parameter optimization, sequential decision-making, reinforcement learning, network optimization, reliability-aware control, and data-driven decision support.
Robust & Trustworthy Learning: domain adaptation, transfer learning, distribution-shift evaluation, sparse-data learning, robustness analysis, model validation, failure-mode assessment, and reliable AI for high-impact systems.
Distributed / Communication-Constrained AI: learning and optimization for wireless and edge systems, network-aware AI, mobility, radio access networks, measurement-driven modeling, and communication-constrained intelligent systems.
Scientific & Domain-Informed ML: 3GPP-informed modeling, digital twins, propagation modeling, generative data augmentation, simulation-driven learning, statistical validation, and physics/domain-informed priors.
Machine Learning Methods: deep learning, reinforcement learning, GANs, autoencoders, classification, regression, anomaly detection, time-series analysis, cross-validation, and experimental evaluation.
Research Tools: Python, PyTorch, TensorFlow, scikit-learn, XGBoost, Pandas, NumPy, SQL, statistical analysis, reproducible experiments, and technical visualization.

Research & Professional Experience

Graduate Research Assistant **09/2019 – 07/2025**
AI4Networks, University of Oklahoma *Tulsa, OK*

- Led research on AI-enabled wireless networks, propagation modeling, digital twins, outage management, RAN optimization, self-healing systems, and wearable intelligence.
- Developed transfer-learning, domain-adaptation, adversarial-learning, generative-modeling, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain problems.
- Built robustness-evaluation frameworks for models operating under distribution shift, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation.
- Designed reusable Python simulation and data-generation platforms, including 3GPP-compliant network simulation and an AI-enabled 5G experimental testbed.
- Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, reproducibility, and technical writing.

Machine Learning Engineer **02/2025 – Present**
AT&T, contractor via Innovatix Inc. *Bedminster, NJ*

- Translate machine-learning research into operational telecommunications systems, emphasizing domain adaptation, reliable model evaluation, and decision support for critical network-assurance workflows.
- Design domain-adaptive neural models and reusable preprocessing pipelines that improve generalization across heterogeneous operational domains and distribution shifts.
- Build cross-domain validation, error-analysis, and failure-mode assessment workflows and collaborate with network engineers, software teams, and public-safety stakeholders on deployable AI systems.

Advanced-Degrees Intern **06/2024 – 08/2024**
AT&T Labs *Austin, TX*

- Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%.
- Investigated 5G NR-NTN cell-selection behavior and improved decision algorithms through data-driven analysis, experimentation, and validation.

Advanced Wireless Technology Development Intern **09/2022 – 01/2023**
Futurewei Technologies *Schaumburg, IL*

- Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets.
- Conducted 50+ experiments to evaluate robustness and generalization under changing data conditions and translated modeling results into recommendations for next-generation communication systems.

Education

PhD, Electrical and Computer Engineering *09/2019 – 07/2025*
University of Oklahoma, Norman, Oklahoma, USA

MSc, Telecommunication Engineering *08/2014 – 08/2016*
University of Engineering and Technology, Taxila, Pakistan

BSc, Telecommunication Engineering *11/2010 – 06/2014*
University of Engineering and Technology, Taxila, Pakistan

Selected Publications

On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks *2025*
Sensors — data-driven multi-parameter optimization, proactive reliability, and decision support for future cellular systems.

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin *2024*
IEEE PIMRC — robust data-driven propagation modeling, generative learning, and resilience under scarce measured data and distribution shift.

Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks *2024*
IEEE SmartNets — machine-learning-based outage detection, intelligent management, and reliability in challenging radio environments.

Towards Positioning Error Impact Characterization and Minimization in User-Centric RAN *2022*
IEEE WCNC — data-driven characterization and optimization for user-centric radio access networks.

Toward a Hybrid RF/Optical Lunar Communication System (LunarComm) *2022*
IEEE Network — architecture and design tradeoffs for resilient, high-data-rate hybrid networked systems.

Unification of RF Energy Harvesting Schemes under Mixed Rayleigh–Rician Fading Channels *2020*
AEÜ – International Journal of Electronics and Communications — analytical modeling of energy-harvesting cooperative-relaying systems under mixed fading conditions.

Real-World Medication Adherence Monitoring Using Hybrid Neural Networks and Smartwatches *2026*
HealthINF (Accepted) — deep learning, multimodal wearable sensing, cross-user evaluation, and reliable human-centered AI.

Selected Research Projects

Multi-Parameter Optimization, Reliability & Self-Healing for 5G/6G *Research Program | 2020–2025*
Developed data-driven methods for outage management, proactive reliability, network optimization, and autonomous decision support in cellular systems. The work connects learning-based models with sequential network decisions and reliability objectives, including robustness across changing radio environments.
Methods & tools: reinforcement learning, network optimization, reliability analysis, outage management, deep learning, statistical modeling, Python, model validation, 5G/6G systems.

Propagation Models & Wireless Digital Twins *Research Project | 2024*
Developed robust propagation models for wireless digital twins using domain-informed generative learning and scarce measured data. Combined 3GPP-informed pretraining, GAN-based augmentation, domain adaptation, and statistical validation to improve resilience under distribution shift.
Methods & tools: propagation modeling, digital twins, GANs, domain adaptation, 3GPP models, synthetic data, statistical validation, robustness evaluation, Python.

SyntheticNET: 3GPP-Compliant Network Simulation *Research Project | 2020–2024*
Developed a Python-based, 3GPP-aligned simulation environment for realistic mobility, propagation, and cellular-network experimentation when measured data is sparse or costly. Enabled controlled data generation and repeatable studies in optimization, digital twins, and resilient wireless AI.
Methods & tools: Python, 3GPP, wireless simulation, mobility modeling, propagation modeling, synthetic data, experiment design, network analytics.

TurboRAN Experimental Platform *Research Infrastructure | 2020–2024*
Built and used a multi-band, multi-tier experimental platform for evaluating AI-enabled radio-access-network methods under controlled conditions. Supported repeatable network measurements, testbed experimentation, RAN optimization studies, and validation of emerging algorithms.
Methods & tools: 5G testbeds, RAN, network measurement, experiment design, wireless prototyping, performance evaluation, optimization.

Teaching & Mentoring

Lecturer **Prior Academic Appointment**

Engineering and Technology Programs

- Taught and mentored undergraduate engineering students in communications, programming, and foundational engineering subjects.
- Emphasized engineering problem solving, implementation, quantitative reasoning, and connection of theoretical concepts to practical systems.
- At the University of Oklahoma, mentored junior researchers in machine-learning implementation, experimental design, reproducible analysis, technical writing, and research presentation.
- Research and teaching interests include machine learning, optimization and decision-making, wireless communications, data science, simulation, and AI-enabled engineering systems.

Teaching Areas & ISE/ECE Contribution

- Prepared to contribute to undergraduate instruction in programming, machine learning, data science, signals and systems, wireless communications, and engineering experimentation, connecting mathematical foundations with implementation and real-world systems.
- Graduate teaching interests include machine learning for engineers, optimization and AI-enabled decision-making, reliable and robust ML, reinforcement learning, networked intelligent systems, and data-driven modeling for cyber-physical and communication systems.
- Favor project-based learning that integrates problem formulation, reproducible implementation, quantitative evaluation, technical communication, and critical comparison of modeling or optimization choices.
- Interested in mentoring ISE and ECE students across methodological and application boundaries, including joint projects linking optimization, ML, communication constraints, digital twins, and trustworthy autonomous systems.

Research Trajectory for Optimization for Machine Learning

My forward research program centers on optimization and learning for reliable networked intelligent systems: (1) robust and adaptive optimization for ML under distribution shift, uncertainty, and limited labels; (2) sequential and hierarchical decision-making for autonomous network control and AI-enabled systems; (3) distributed and communication-efficient learning and optimization for wireless/edge environments; and (4) trustworthy AI optimization emphasizing reliability, robustness, interpretability, and rigorous evaluation. This agenda builds directly on my prior work in multi-parameter network optimization, reinforcement learning, domain adaptation, digital twins, outage management, and experimentally validated wireless AI.

Collaboration, Research Translation & Academic Service

- Collaborate across academic research, telecommunications R&D, software engineering, public-safety operations, and domain specialists to translate complex problems into measurable research and engineering objectives.
- Bring industry R&D experience from AT&T Labs and Futurewei together with university research, providing a foundation for interdisciplinary partnerships, externally relevant research questions, and translational projects.
- Maintain reproducible research workflows, careful validation, professional technical communication, and a collegial approach to mentoring, collaboration, peer review, and knowledge sharing.